(Please write your Enrollment Number)

Enrollment No._____

MID-TERM EXAMINATION

Course Name : B.Tech CSE/ECE-AI/ECE/IT/MAE/DMAM/Reappear Sem : IV

(March, 2024) OFF LINE mode

Subject Code: BCS 202	Subject: Computer Organization and Architecture

Time : 1 ½ Hours Maximum Marks : 30

Attempt all questions

Q1	Attempt all parts	(2.5*4)
	a. Realize the XOR gate by using NAND or NOR Gate only and show its truth table for two inputs XOR Gate	
	b. Explain the functionality of the D flip flop with the help of its truth table. Justify how D F/f can be used as a memory element?	
	c. Calculate 2's complement of the following numbers in 8 bits format and represent it in IEEE-754 floating point notation (i) 10010011 and (ii) 10101110	
	d. What are the micro-operations and explain the microoperations for any instruction with the example	
Q2	**Attempt any two parts [UNIT1-]**	(5,5)
	(a) Simplify and then Realize the given following Boolean equation using basic gates ABC+AC+ C+D and find how many minimum numbers of gates are required?	
	(b) What are the different methods that are used to represent the numbers in computers and explain any two methos that you suggest along with its merits and demerits?	
	(c) What is a bus and what are the different buses being used in computers organization and explain its functions with the help of block diagram?	
		(5,5)
Q3	**Attempt any two parts [UNIT2-]**	
	(a) What is an instruction and explain instruction with the help of instruction cycle considering the **MOV A, #30h** [Move the data 30 into Accumulator]	
	(b) Design a 4-bit parallel adder and subtractor and how many minimum full adders and half adders are required.	
	(c) What is microprogrammed control unit. How an instruction sequence is generated using microprogram. Define the functioning of control registers.	